

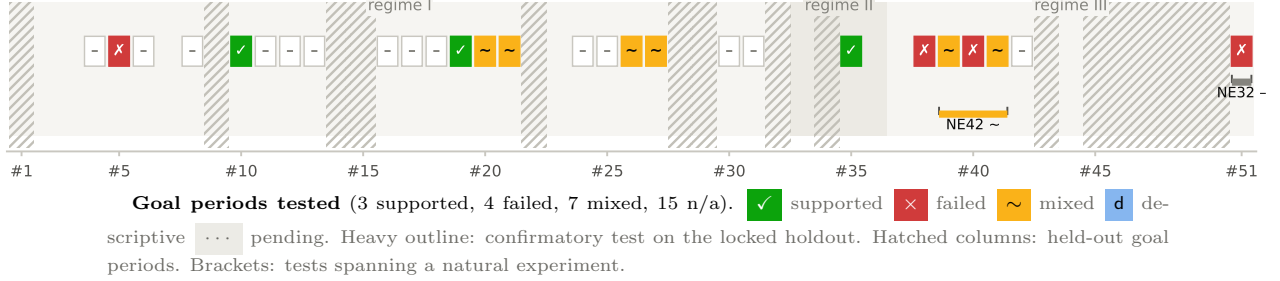
H48: The settling time is a mixing time on the read-out graph

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Agents read each other within minutes but content settles over hours; the noise is too high to test scaling with a slower read-out time.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim is that content settles after a kickoff once every agent has read every other agent. The settling time is then a bulk mixing time on the read-out graph, not whole-graph λ_2 . The DQ1 context ledger gives two read-out times per period. T_{90} is the time until 90% of room-mate pairs have read each other; t_{mix} is the mixing time of the read-out chain. The test compares both with the kickoff-remnance settling time τ (1-h bins, two embedding models) on 27 non-holdout periods. Native tests: NE42, #38, #51 joins, NE32. Coverage takes minutes (T_{90} median 0.24 active h). Settling takes hours (τ median 4.5 h, about $10\times$). Neither predictor beats a constant (LOPO gains -0.01 to -0.14). But τ is detectable in only 11/27 periods, and a noise-calibrated synthetic has no power. **Inconclusive**; the shallow-coverage version fails on magnitude. The holdout script exists and is not run.

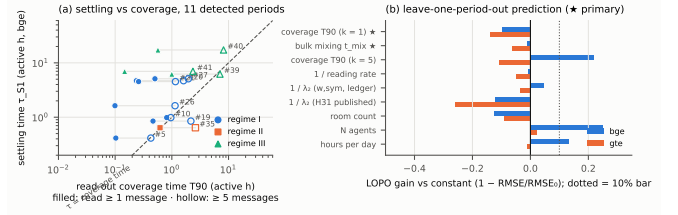
PHYSICAL MODEL

Vector spins (model 11) that update only at read-out calls. Agent i 's content $x_i \in \mathbb{R}^{32}$ is its whitened statement mean. The kickoff quenches it toward the target $\hat{k}$ (H54). At each call, x_i moves toward the messages that newly entered its context (ledger batch B), with weight α . W_{ij} counts j 's messages read by i per active hour; u_i is the reading-turn rate; Q is the read-out chain. Rivals: a field clock, the own-call clock, the day clock and λ_2 (H31).

$$x_i \leftarrow (1 - \alpha) x_i + \alpha \langle x_{s(m)} \rangle_{m \in B}, \quad Q_{ij} = u_i \frac{W_{ij}}{\sum_k W_{ik}} - \delta_{ij} u_i$$

$$C_k(t) = \Pr_{(i,j)} [n_{ij}(t) \geq k], \quad t_{\text{mix}}^{\text{bulk}} = \text{med}_i \min \{t : \text{TV}(e_i e^{Q^t}, e_i e^{Q^\infty}) \leq \frac{1}{4}\}$$

$$A_{\text{ex}}(t) = \langle \cos(v_i, \hat{k}) \rangle_i - \langle \cos(v_i, \hat{k}_q) \rangle_{i,q} = A_\infty + (A_0 - A_\infty) e^{-t/\tau}$$



Observables. (a) Settling time against coverage time for the 11 periods with detected settling. Filled: 90% of pairs read once. Hollow: read five times. Settling sits about $10\times$ above single-read coverage (dashed line $\tau = T_{90}$) and near depth-5 coverage (post hoc). (b) Leave-one-period-out gain over a constant. Both primaries (stars) lose in both models. The bge-only leads (depth-5 coverage, N) do not replicate with gte.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Reading is not the bottleneck in broadcast rooms. Every agent reads every room-mate within minutes of a kickoff (T_{90} 0.24 active h), yet content relaxes for hours. Do not merge rooms or raise call rates to speed convergence on a new goal. If read-out drives settling at all, each read moves an agent by about $1/250$. Coverage times from the ledger are cheap, but one silent agent sets a whole room's coverage. The scaling question stays open: per-period settling times are too noisy and too varied in shape to rank 27 periods.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 T_{90} or t_{mix} beats constant, both models	gains $-0.10/-0.14, -0.01/-0.06$; 11 detected (< 12)	inconclusive
P2 $\tau/T_{90} \in [1/3, 3]$	10.1 (IQR 2.9–18.6); gte 4.5	failed
P3 bulk beats worst case	0/3 pairs	failed
P4 holds within regime	gains $-0.08, -0.12$	failed
P5, P6 robust to model, DQ5	ρ 0.80, 0.97	held
P8 settling detected $\geq 60\%$	41% bge, 52% gte	failed
P10 no λ_2 beats constant	best gain 0.05	held
NE42 τ ordered as T_{90}	ρ 0.30 bge, 0.90 gte (n 5)	mixed
#38 slow-covering room settles later	settles faster (gte τ_q 1.6 vs 12.3 d)	failed
#51 newcomers assimilate	3/11; they individuate	failed

GOAL PERIODS IN DETAIL

G05 $\times$ failed
G10 $\checkmark$ supported
G19 $\checkmark$ supported
G20 $\sim$ mixed
G21 $\sim$ mixed
G26 $\sim$ mixed
G27 $\sim$ mixed
G35 $\checkmark$ supported
G38 $\times$ failed
G39 $\sim$ mixed
+19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

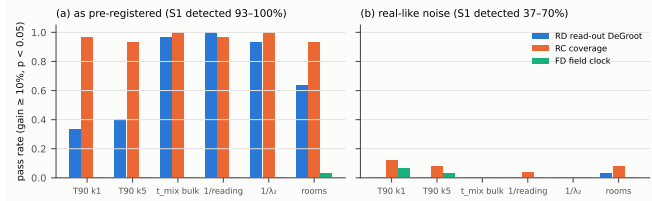
The settling estimate is fragile. A single exponential fits in about half the periods. The rest show a fast drop then recovery, multi-day declines, or no kickoff signal (#51's private goals). H20's day-scale τ_q exists only in 10 long periods. The bge and gte detected sets overlap in 9 of 16 periods. Coverage counts chat messages only, not artifacts. Regime-I/II read times use low-confidence call starts. The analysis looks at 19 predictors $\times$ 2 models; only P1 has multiplicity control, so the bge leads are not tests. The real-like calibration comes after the data.

ROUND 2

Estimate the per-read pull α at message level (H29-style, on ledger read-outs) and predict $\tau = 1/(\alpha u)$ directly. Build a hierarchical settling estimator that recovers a planted τ at real noise. Test depth-5 coverage on the holdout.

- **H48-R1** Message-level α .
- **H48-R2** A settling estimator with power.
- **H48-R3** Depth-5 coverage (post hoc lead).
- **H48-R4** Newcomer individuation (from G51).

H48-settling-mixing-time · generated 2026-10-04 11:52 by infra/summaries/build_summaries.py · card hypotheses/H48-settling-mixing-time/README.md · ratings: H48 agent (suggested) (page 2)



Second figure. Synthetic: vector spins on the real read-out schedules. (a) At the pre-registered noise level, bulk predictors identify read-out truths (RD, RC) with no false positives under a field clock (FD). (b) At noise matched to the real detection rate (post hoc), no predictor passes under any truth. The real-data null therefore says nothing about scaling.